

A 3.3.1 Data Language Types



A 3.3.1 Outline the differences between data language types within SQL.

Data Definition Language (DDL) and Data Manipulation Language (DML) are two distinct but equally important categories of SQL commands. DDL is concerned with defining and managing the structure of the database, while DML is focused on working with the actual data stored within that structure.

Database Programming with SQL - Understanding Data Language Types

As you learn about databases (Theme A, A3), you'll discover that Structured Query Language (SQL) is the fundamental language used to interact with them. This page will focus on Data Language Types, outlining the differences between two types within SQL.

- Data Definition Language (DDL)
- Data Manipulation Language (DML).

Understanding these two categories is crucial for effectively defining and working with data in relational databases.

Data Definition Language (DDL)

Data Definition Language (DDL) is the set of SQL commands used to define the structure or schema of a database. Think of DDL as the blueprint for your database. It allows you to create, modify, and delete database objects such as tables, indexes, and views.

DDL focuses on the organisation and framework of the data rather than the data itself. Here are some key DDL statements you need to be familiar with:

CREATE TABLE

—

This command is used to create a new table in the database. When creating a table, you need to define its name and the columns it will contain. For each column, you specify its name and the data type it will hold (e.g., INTEGER, TEXT, REAL, DATE). You can also define constraints, such as primary keys, foreign keys, and uniqueness, which are crucial for ensuring data integrity.

Example: Let's say you want to create a table called **Customers** to store customer information.

```
CREATE TABLE Customers (  
    CustomerID INTEGER PRIMARY KEY,  
    FirstName TEXT NOT NULL,  
    LastName TEXT NOT NULL,  
    Email TEXT UNIQUE,  
    RegistrationDate DATE  
);
```

In this example:

- **CREATE TABLE Customers** initiates the creation of a table named **Customers**.
- **CustomerID INTEGER PRIMARY KEY** defines a column named **CustomerID** which will store integer values and will serve as the primary key for this table, ensuring unique identification of each customer.
- **FirstName TEXT NOT NULL** defines a column **FirstName** to store text data, and **NOT NULL** constraint ensures that this field cannot be left empty.
- **LastName TEXT NOT NULL** similarly defines a non-empty **LastName** column for text data.
- **Email TEXT UNIQUE** defines an **Email** column for text data, and the **UNIQUE** constraint ensures that no two customers can have the same email address.
- **RegistrationDate DATE** defines a column to store the date of registration.

ALTER TABLE

This command is used to modify the structure of an existing table. You can use it to add new columns, delete existing columns, modify the data type of a column, or add or remove constraints.

Example: Suppose you want to add a **PhoneNumber** column to the **Customers** table.

```
ALTER TABLE Customers  
ADD PhoneNumber TEXT;
```

This statement adds a new column named **PhoneNumber** with the data type TEXT to the **Customers** table.

DROP TABLE

This command is used to delete an entire table from the database. This action permanently removes the table and all the data it contains, so it should be used with caution.

Example: To delete the **Customers** table:

```
DROP TABLE Customers;
```

DDL statements are crucial in the database design process (A3.2). When you describe database schemas (conceptual, logical, physical) and construct Entity Relationship Diagrams (ERDs), the DDL commands are what translate these designs into actual database structures.

The choice of data types and the implementation of primary and foreign keys directly using DDL ensure data integrity and facilitate relationships between tables.

Understanding normal forms (1NF, 2NF, 3NF) and applying them often involves using DDL to structure tables appropriately, addressing issues like data duplication and dependencies.

Data Manipulation Language (DML)

Data Manipulation Language (DML) is the set of SQL commands used to manipulate the data within the database structures defined by DDL. DML allows you to insert new data, retrieve existing data, update data, and delete data.

Here are the primary DML statements you need to understand:

SELECT

This is perhaps the most frequently used DML command. It is used to retrieve data from one or more tables. **SELECT** statements allow you to specify which columns you want to see, filter the rows based on certain conditions (**WHERE**), sort the results (**ORDER BY**), group data (**GROUP BY**), and perform calculations using aggregate functions (**AVERAGE**, **COUNT**, **MAX**, **MIN**, **SUM** - HL only). You can also perform joins to combine data from two or more related tables.

Example: To retrieve the first and last names of all customers:

```
SELECT FirstName, LastName  
FROM Customers;
```

Example (with filtering): To retrieve the first and last names of customers registered after a specific date:

```
SELECT FirstName, LastName  
FROM Customers  
WHERE RegistrationDate > '2024-01-01';
```

Example (with join): If you have an **Orders** table with a **CustomerID** as a foreign key, you can retrieve customer names along with their order details:

```
SELECT c.FirstName, c.LastName, o.OrderID, o.OrderDate  
FROM Customers c  
JOIN Orders o ON c.CustomerID = o.CustomerID;
```

INSERT INTO

This command is used to add new records (rows) into a table. You need to specify the table name and the values for each column (or a subset of columns).

Example: To insert a new customer into the **Customers** table:

```
INSERT INTO Customers (CustomerID, FirstName, LastName, Email, RegistrationDate)
VALUES (101, 'Alice', 'Smith', 'alice.smith@example.com', '2024-03-15');
```

UPDATE SET

This command is used to modify existing records in a table. You need to specify the table name, the columns you want to update and their new values (**SET**), and typically a condition (**WHERE**) to identify which rows should be updated.

Example: To update the email address of the customer with **CustomerID** 101:

```
UPDATE Customers
SET Email = 'alicesmith@newemail.com'
WHERE CustomerID = 101;
```

DELETE FROM

This command is used to remove records from a table. You specify the table name and usually a condition (**WHERE**) to indicate which rows should be deleted. If no **WHERE** clause is provided, all rows in the table will be deleted (use with extreme caution!).

Example: To delete the customer with **CustomerID** 101:

```
DELETE FROM Customers
WHERE CustomerID = 101;
```

DML statements are essential for the day-to-day operations of a database. They allow users and applications to interact with the stored data to perform tasks such as displaying information, processing transactions, and generating reports.

Understanding how to construct effective **SELECT** queries, including those involving joins, relational operators, filtering, pattern matching, and ordering, is a key skill in database programming. Furthermore, knowing how to update data using **INSERT**, **UPDATE**, and **DELETE** is fundamental for maintaining the accuracy and relevance of the information stored in the database.

Key Differences Summarised

To clearly understand the distinction between DDL and DML, consider the following:

Feature	Data Definition Language (DDL)	Data Manipulation Language (DML)
Purpose	Defines the database schema (structure)	Manipulates the data within the schema
Operations	Create, alter, delete database objects	Insert, retrieve, update, delete data rows
Objects Acted Upon	Tables, indexes, views, databases, etc.	Rows (records) within tables
Keywords (Examples)	CREATE , ALTER , DROP	SELECT , INSERT INTO , UPDATE , DELETE

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